

Healthcare Analytics - Week 6

Power BI Report: Patient, Admission and Billing Dashboard

Tool used: Microsoft Power BI Desktop

1. Introduction

This week I built a healthcare analytics dashboard in Power BI. I used patient, admission and billing data and put them together in one interactive report. The dashboard shows the main KPIs, patient demographics, admission patterns, billing and yearly trends, and I added slicers so the data can be filtered.

2. Objectives

- Build a clear healthcare dashboard in Power BI.
- Show the main patient and billing KPIs.
- Look at patient numbers by age group and gender.
- Compare medical conditions with admission types.
- Show how billing and patients change over the years.
- Add slicers for insurance provider, admission type and admission date.

3. Tools Used

I used Microsoft Power BI Desktop for everything in this project: loading and modelling the data, writing the DAX measures, making the visuals and setting up the filters. The KPI cards are built from measures I created in the Billing table.

4. Data Model and Fields

The model has four tables: Billing, Dim_Admission, Dim_Patient and Healthcare_Data. The fields I mainly worked with were Admission Type, Age, Billing Amount, Date of Admission, Discharge Date, Gender, Hospital, Insurance Provider, Medical Condition, Name and Patient ID. I also created a few measures on top of these.

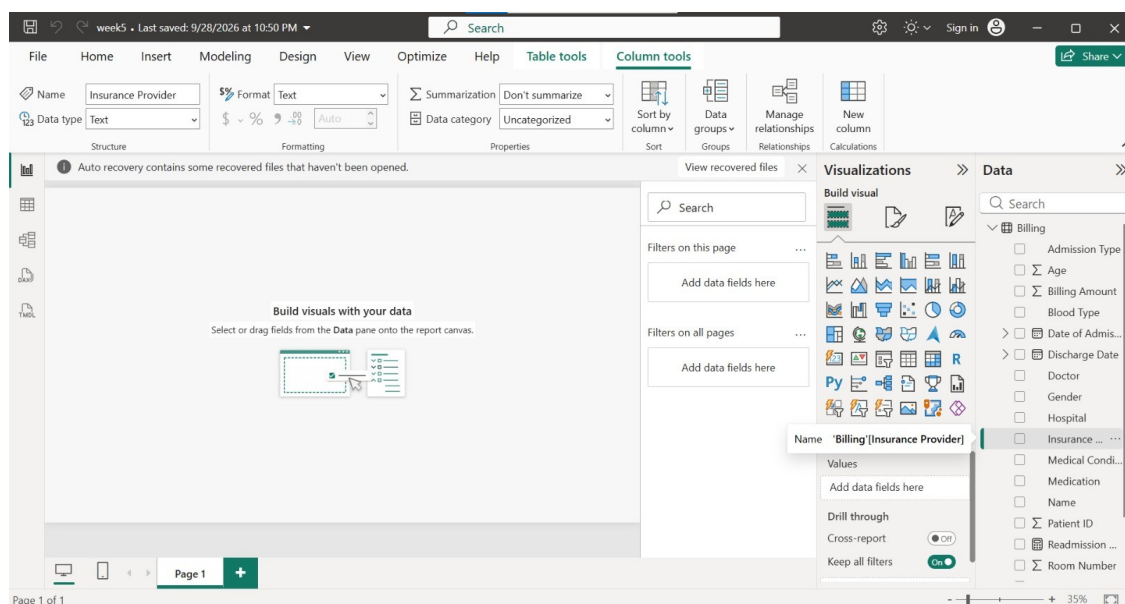


Figure 4.1: Fields and data setup in Power BI

Before adding the KPIs I first arranged a basic layout for the page so I knew where each visual would go.

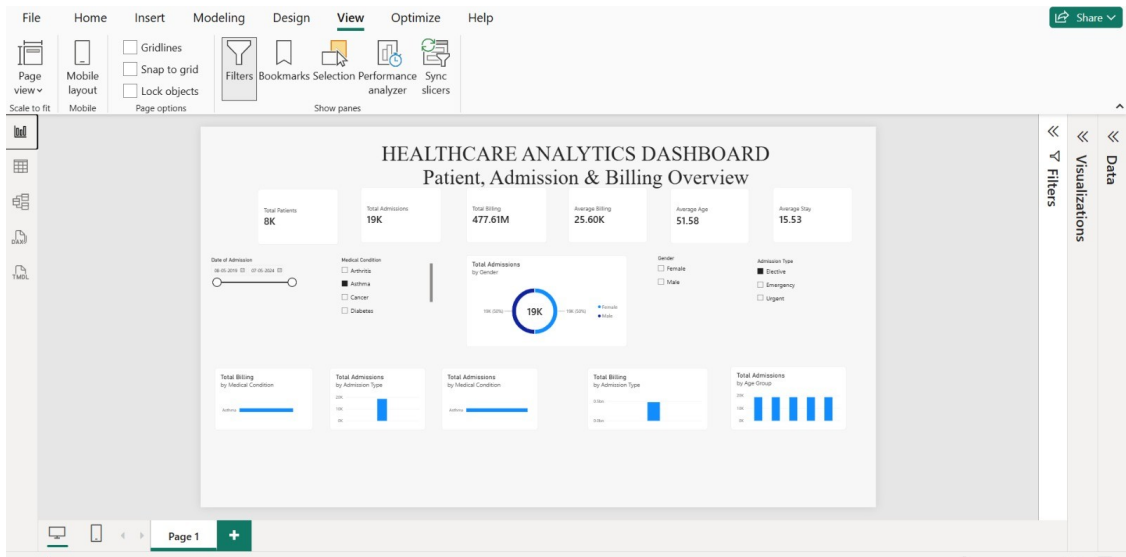


Figure 4.2: First draft of the dashboard layout

5. KPI Cards

I made cards for Total Patients, Total Billing Amount, Average Billing Amount and Average Length of Stay. While building, I also tried cards for Total Admissions and Average Age, but the final page uses the four main ones.

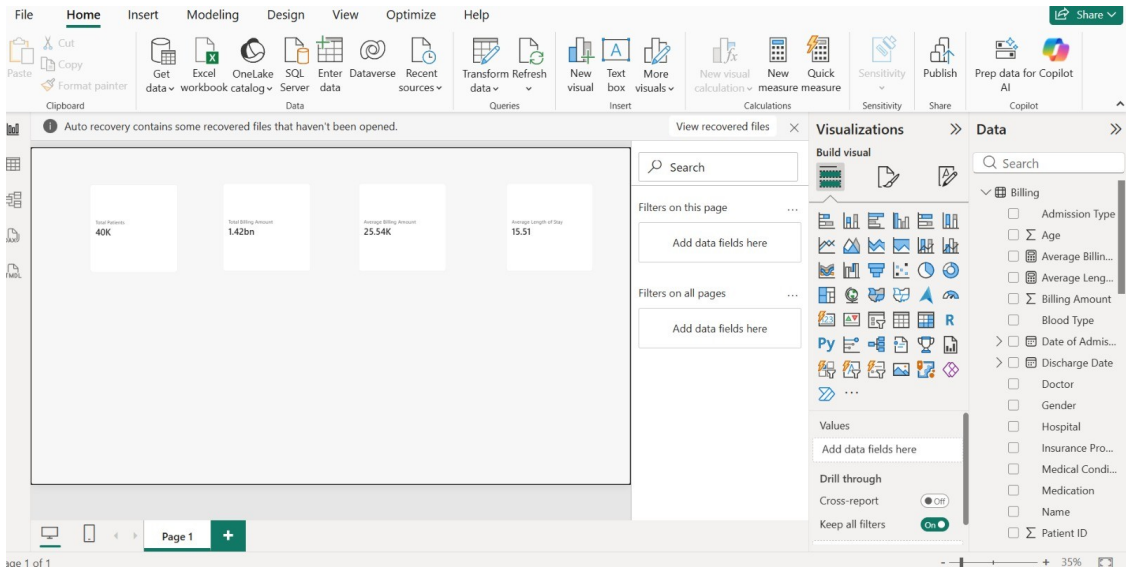


Figure 5.1: KPI cards

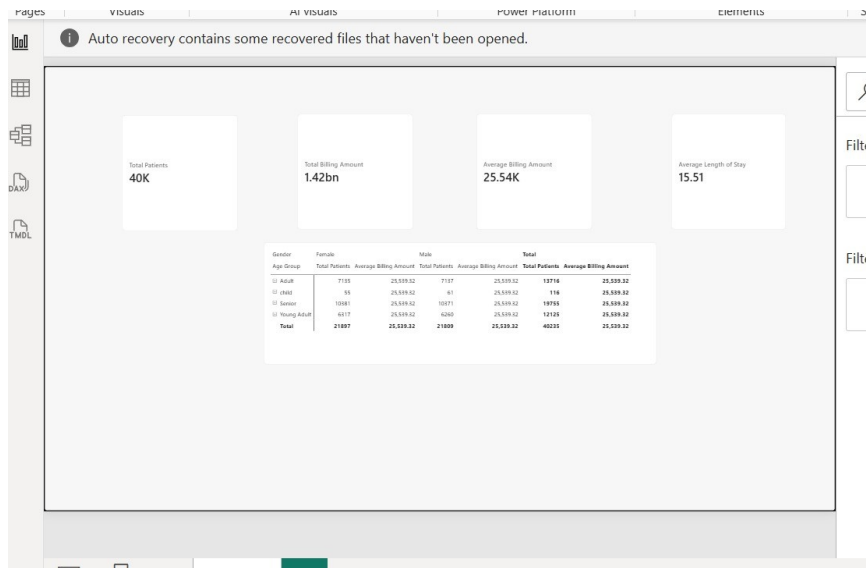


Figure 5.2: KPI cards with the demographic table

6. Patient Demographics

I used a matrix to compare the number of patients and the average billing amount across age groups and gender. Next to it I added a donut chart that shows what share of patients falls in each age group.

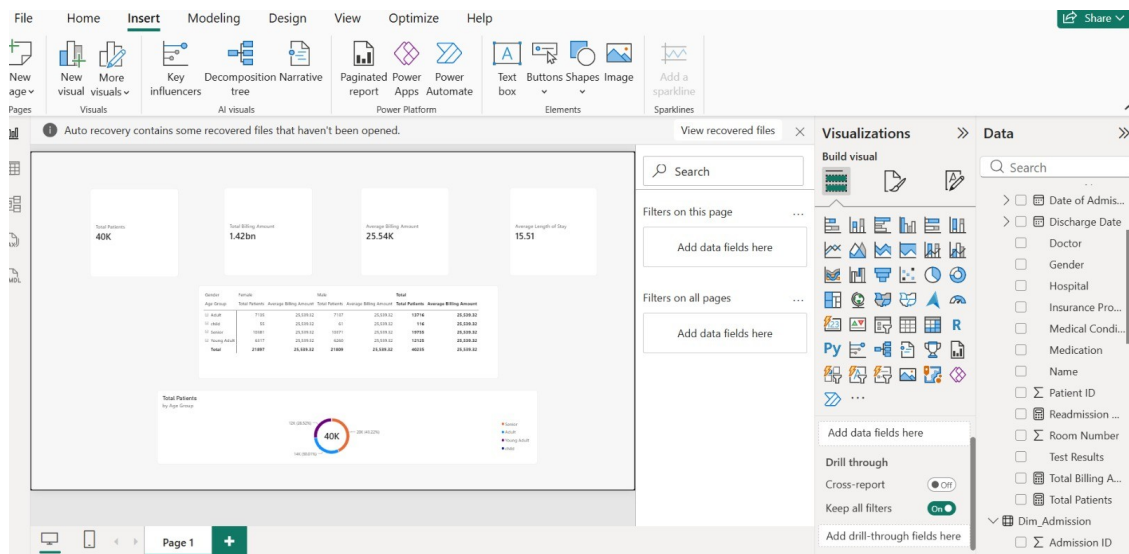


Figure 6.1: Age group and gender analysis

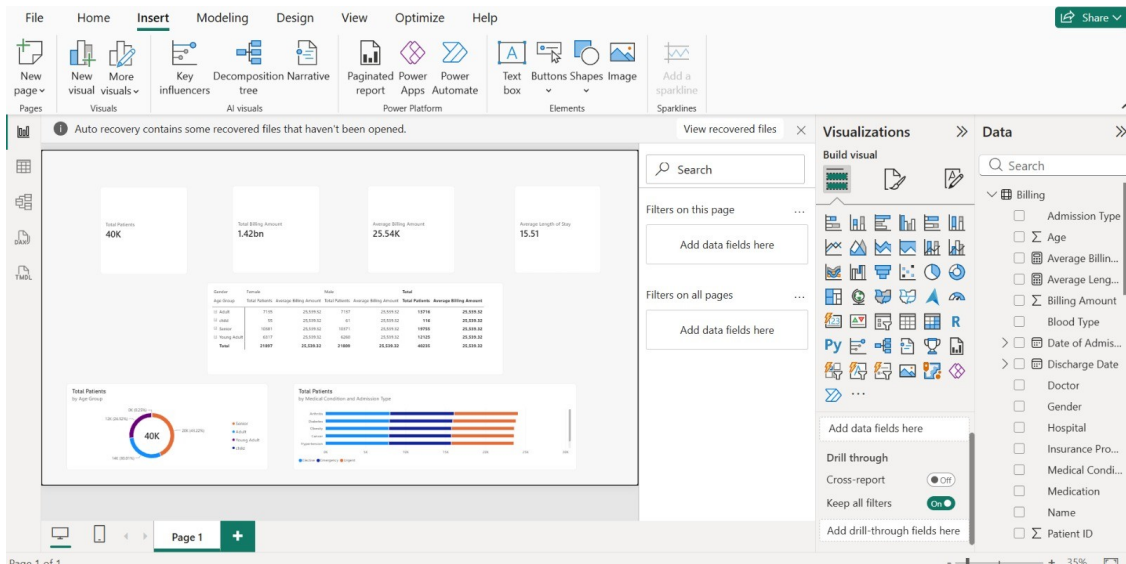


Figure 6.2: Demographic visualizations added to the dashboard

7. Medical Condition and Admission Type

To compare admission types across conditions I used a stacked bar chart of Total Patients by Medical Condition and Admission Type. The three admission types are Elective, Emergency and Urgent, so it is easy to see how each condition is split.

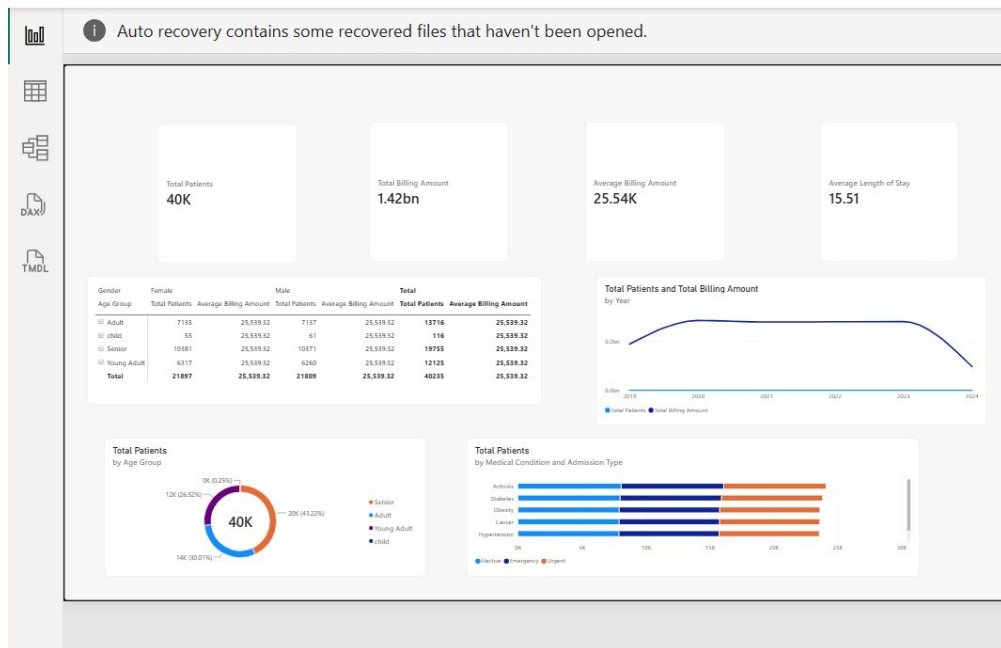


Figure 7.1: Medical condition and admission type chart

8. Trend by Year

The line chart "Total Patients and Total Billing Amount by Year" shows how both values change over the years in the data.

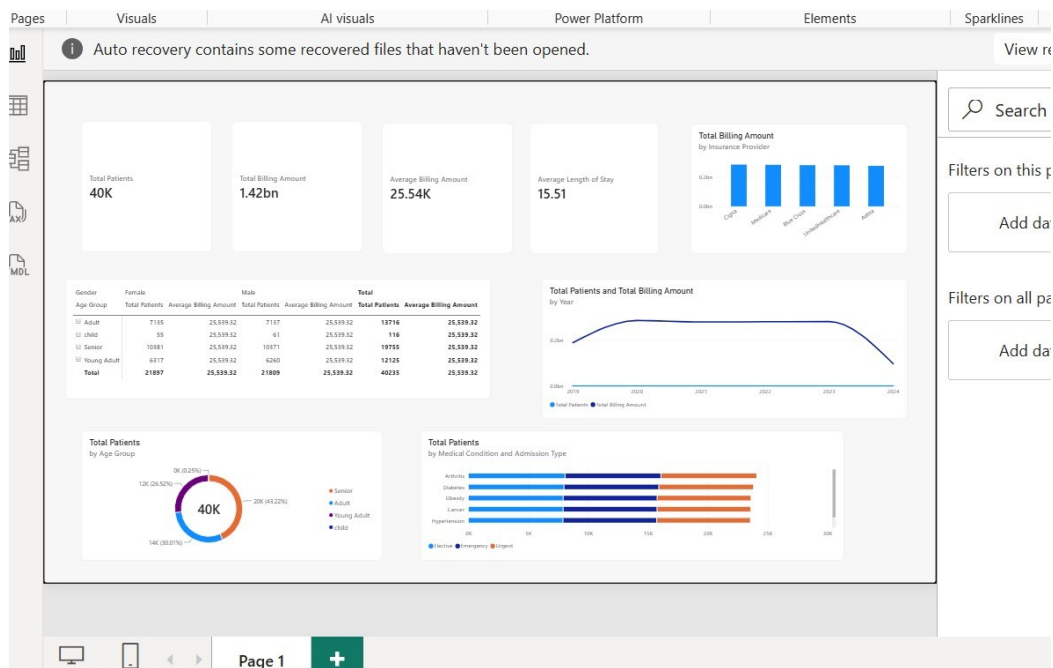


Figure 8.1: Patient and billing trend by year

9. Billing by Insurance Provider

A column chart shows Total Billing Amount for each insurance provider. The providers in the data are Aetna, Blue Cross, Cigna, Medicare and UnitedHealthcare.

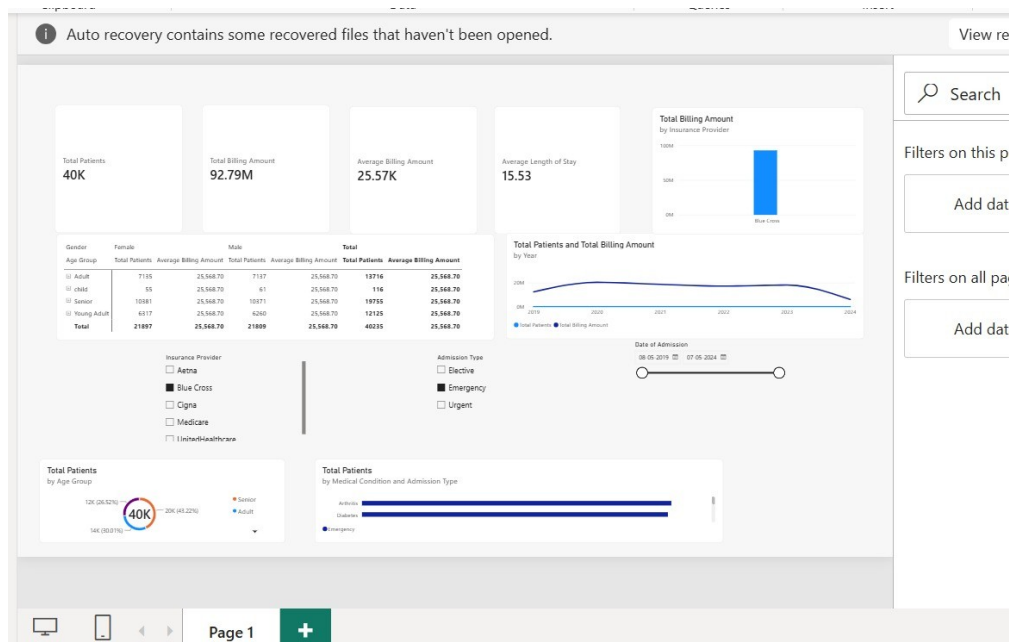


Figure 9.1: Billing by insurance provider

10. Slicers

I added slicers for Insurance Provider, Admission Type and Date of Admission. When a value is selected, the other visuals on the page change to match it.

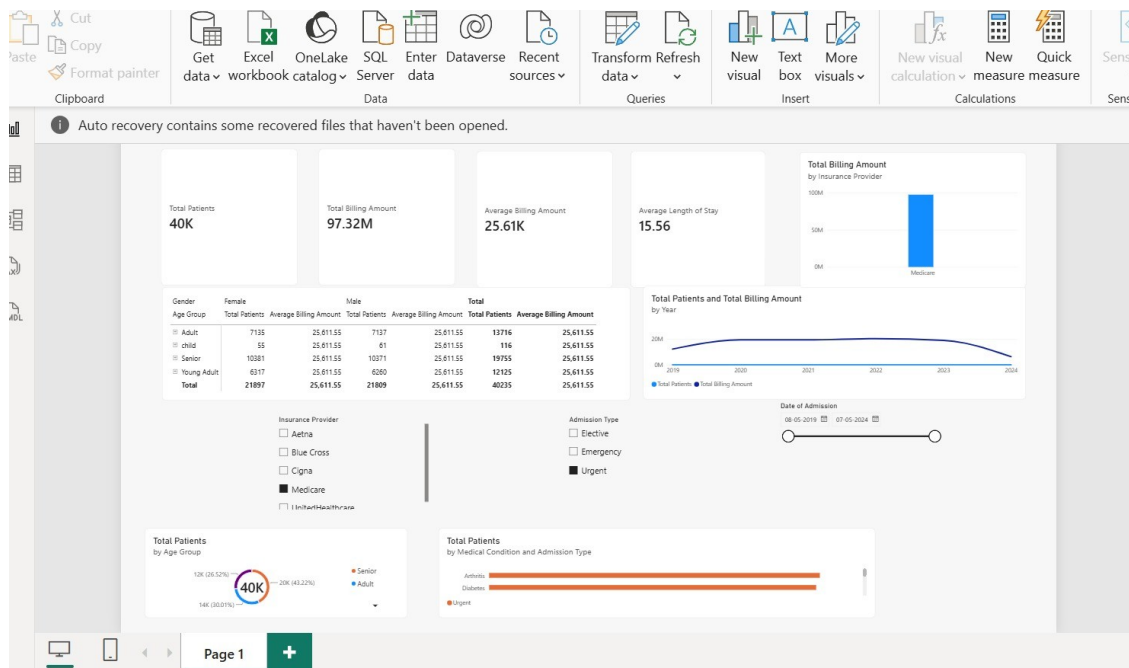


Figure 10.1: Slicers used for filtering

11. Testing the Filters

I tried different selections on the slicers, such as one insurance provider, one admission type and different date ranges, to check that everything updates properly. The KPI cards, table, trend chart and the other visuals all changed with the selection, so the filtering works as expected.

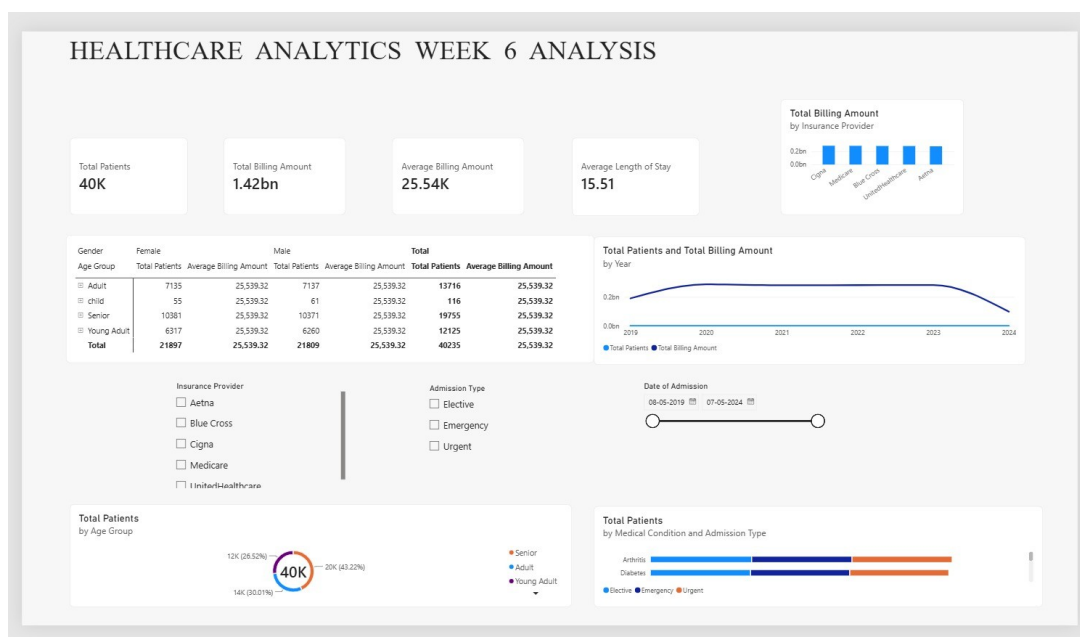


Figure 11.1: Dashboard after filtering

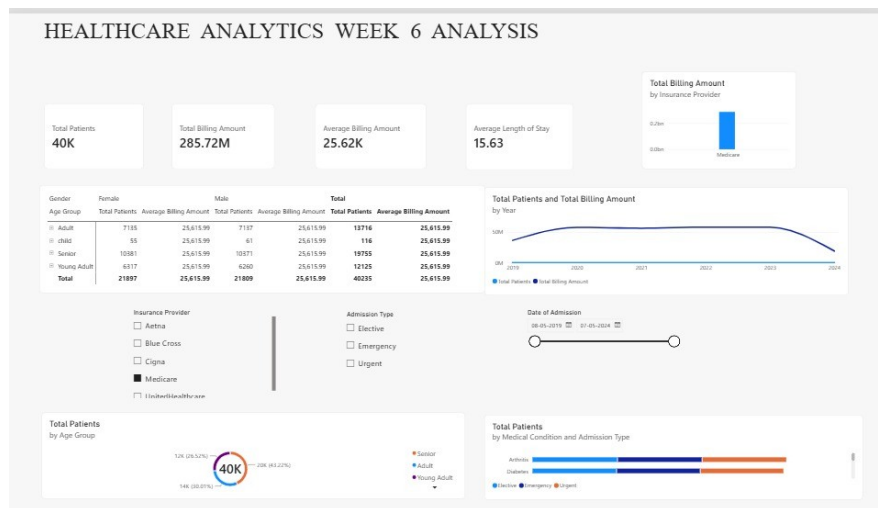


Figure 11.2: Dashboard with a slicer selected

12. Final Dashboard

The final dashboard has the KPI cards, demographic matrix, insurance billing chart, yearly trend, slicers, age group donut and the condition vs admission type chart, all on a single page.

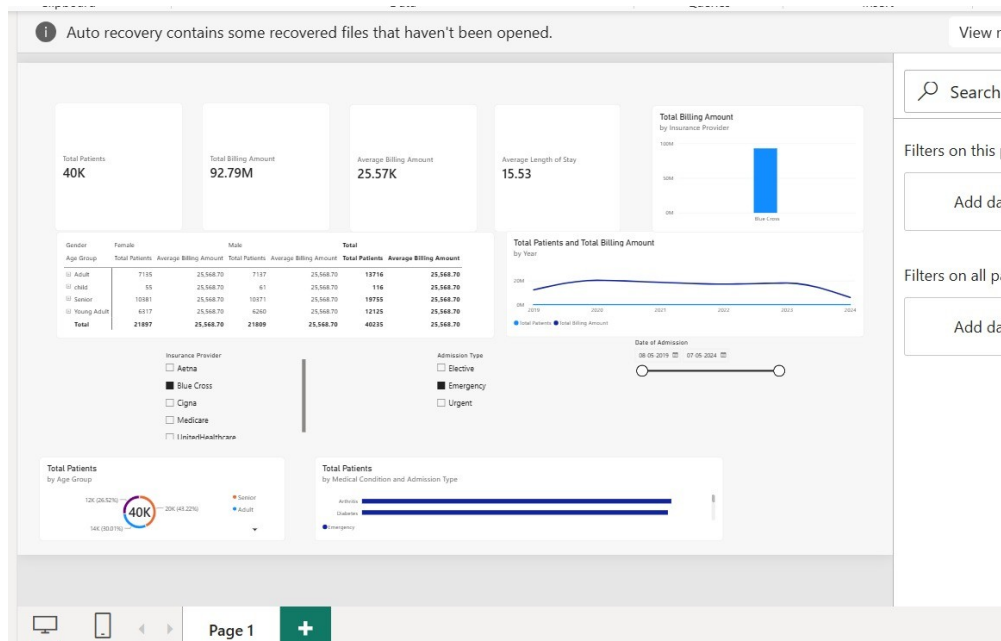


Figure 12.1: Final dashboard layout

13. Observations

The dashboard gives a quick overall picture of patient count, billing and length of stay. The matrix shows how patients are spread across age groups and gender, and the donut chart makes the age group share easy to see. The stacked bar chart helps compare admission types for each condition, the insurance chart compares billing between providers, and the line chart shows the yearly trend.

14. Conclusion

In Week 6 I turned the Power BI project into an interactive dashboard. I used measures, different visuals, slicers and filters, and arranged them on one page so the healthcare data is easy to read. The screenshots above show the steps I followed and the final output.